

Human Resource Management System (HRMS)

Requirement Analysis

The Human Resource Management System (HRMS) is designed to streamline HR operations by managing employee records, payroll, attendance, recruitment, and performance. Below is a detailed requirement analysis of the HRMS system, focusing on the entities and modules.

Entities

1. Employee

- Stores details about employees such as personal information, role, department, and contract information.

2. Department

- Represents the different departments in the organization (e.g., HR, IT, Sales).

3. Role

- Describes the roles within departments (e.g., Manager, Developer, HR Executive).

4. LeaveRequest

- Tracks leave requests made by employees, including type, status, and duration.

5. Salary

- Represents the salary details of each employee, including deductions and allowances.

6. Attendance

- Tracks employee attendance, leaves, overtime, and absenteeism.

7. PerformanceReview

- Stores performance reviews, feedback, ratings, and promotion history.

8. Recruitment

- Handles recruitment activities, including job postings and applicant management.

9. Applicant

- Stores data about job applicants, including contact details and the position applied for.

10. JobPosting

- Represents job openings and descriptions for different roles in the organization.

11. Interview

- Tracks interviews scheduled for job applicants, including interview feedback and status.

12. Contract

- Represents employment contracts, including start and end dates, contract type, and terms.

13. Task

- Tracks tasks assigned to employees as part of their role or project.

14. Notification

- Stores notifications for employees, such as task deadlines, leave approvals, or announcements.

15. Payroll

- Manages the payroll process, including salary computation, tax, and benefits.
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Modules

1. Employee Management

- **Functionality:** Manages employee records, including personal details, department assignment, role assignment, and contracts.
- **Key Features:**
 - Add, update, and remove employee records.
 - Assign employees to departments and roles.
 - Track employee contracts and their status (active, terminated).

2. Attendance Tracking

- **Functionality:** Tracks employee attendance and leave requests.
- **Key Features:**
 - Record daily attendance (check-in/check-out).
 - Manage leave requests and approvals.
 - Calculate overtime and absenteeism.
 - Generate attendance reports for employees.

3. Payroll Management

- **Functionality:** Handles salary calculations, deductions, and generation of pay slips.
- **Key Features:**
 - Calculate employee salaries based on attendance and performance.
 - Handle tax deductions, allowances, and bonuses.
 - Generate monthly or bi-weekly pay slips.
 - Manage payroll records and generate payroll reports.

4. Performance Management

- **Functionality:** Manages performance reviews, feedback, and promotions.
- **Key Features:**
 - Schedule and conduct performance reviews.
 - Store review ratings and feedback for employees.
 - Track employee promotions and role changes.
 - Generate performance reports.

5. Recruitment

- **Functionality:** Manages job postings, applicants, and the interview process.

- **Key Features:**

- Create and manage job postings.
- Track applicant information and applications.
- Schedule and track interviews.
- Store interview feedback and hiring status.

6. Leave Management

- **Functionality:** Tracks and manages employee leave requests and approvals.

- **Key Features:**

- Create and submit leave requests.
- Approve or reject leave requests by managers.
- Track leave balances and usage for employees.
- Generate reports on leave statistics.

7. Reporting Module

- **Functionality:** Generates reports on various HR metrics such as employee performance, payroll, and leave statistics.

- **Key Features:**

- Employee performance reports.
- Payroll reports, including salary, deductions, and bonuses.
- Leave and attendance reports.
- Recruitment reports (hiring status, job postings, applicant details).

Sample Use Cases

1. **Employee Onboarding:** HR staff adds a new employee into the system, assigns a department and role, sets the contract details, and generates login credentials.
2. **Attendance Management:** Employees log their attendance, and the system calculates total hours worked, overtime, and absenteeism. If an employee submits a leave request, it goes through an approval workflow.
3. **Payroll Generation:** At the end of the month, the system calculates employee salaries, considering attendance, overtime, and deductions, and generates pay slips for each employee.
4. **Performance Review:** A manager schedules performance reviews for team members, provides feedback, assigns ratings, and suggests promotions or bonuses, which are updated in the employee's record.
5. **Recruitment Process:** The HR team creates job postings, applicants apply for positions, interviews are scheduled, and interview feedback is recorded. Once a candidate is selected, they are added to the system as a new employee.

Non-Functional Requirements

1. **Security:** Ensure employee records, payroll data, and personal information are secured using encryption and authentication mechanisms.
 2. **Scalability:** The system should be able to handle an increasing number of employees, departments, and HR activities.
 3. **Reliability:** Ensure that payroll calculations and other critical processes are accurate and consistently available.
 4. **Performance:** The system should provide quick access to records and reports, especially during high-demand periods like payroll generation or employee performance reviews.
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This detailed analysis covers the entities, modules, and use cases for the Human Resource Management System, providing a foundation for development using Core Java, Collections, and Hibernate.

Modules Description :

1. Employee Management Module

This module handles all employee-related information and tasks, from adding new employees to updating existing employee details.

Key Features:

- **Employee Profiles:** Create, view, update, and delete employee profiles, including personal information, contact details, and professional background.
 - **Department and Role Assignment:** Assign employees to specific departments and roles within the organization (e.g., assigning an employee to the IT department as a Senior Developer).
 - **Contract Management:** Handle the details of employment contracts, such as start and end dates, contract type (full-time, part-time, temporary), and job tenure.
 - **Employee Status:** Track employee status (active, on leave, terminated) and manage their employment lifecycle from onboarding to offboarding.
 - **Document Management:** Store documents related to employees (e.g., contracts, certifications, performance reviews).
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2. Attendance Tracking Module

This module focuses on monitoring and managing employee attendance, leaves, and work hours.

Key Features:

- **Daily Attendance:** Employees can check in/out, and their attendance is recorded in the system.
 - **Leave Management:** Track leaves requested by employees, including vacation, sick leave, or any other type of leave.
 - **Leave Balance Tracking:** Calculate leave balance for employees based on the leave policy and past leave records.
 - **Overtime and Absenteeism:** Track overtime worked by employees and monitor any absenteeism.
 - **Attendance Reports:** Generate attendance reports for individual employees, teams, or departments.
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3. Payroll Management Module

This module is responsible for calculating employee salaries, managing payroll data, and handling salary payments.

Key Features:

- **Salary Calculation:** Automatically calculate salaries based on work hours, attendance, performance bonuses, tax deductions, and other allowances.
- **Pay Slip Generation:** Generate monthly or bi-weekly pay slips with detailed breakdowns of earnings, deductions, and bonuses.
- **Deductions and Bonuses:** Handle tax deductions, health insurance, and bonuses or incentives for employees.

- **Tax Compliance:** Ensure payroll is in compliance with tax regulations and other statutory requirements.
 - **Payroll History:** Maintain records of past salary transactions and pay slips for reference.
 - **Payment Integration:** Integrate with banking systems for automated salary payments to employee accounts.
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4. Performance Management Module

This module enables managers and HR teams to evaluate employee performance and manage feedback and promotions.

Key Features:

- **Performance Reviews:** Conduct periodic performance reviews (monthly, quarterly, annually), rate employees, and provide feedback on their work.
 - **Goal Setting:** Define and assign goals to employees, and track progress toward achieving them.
 - **Promotion and Appraisal Management:** Track employee promotions, salary increases, and role changes based on performance.
 - **Feedback System:** Managers can provide feedback to employees and record it within the system for future reference.
 - **Performance Reports:** Generate reports on employee performance, highlighting top performers or identifying areas for improvement.
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5. Recruitment Management Module

This module is designed to streamline the recruitment process, from job postings to applicant tracking and hiring decisions.

Key Features:

- **Job Postings:** HR teams can create, edit, and manage job listings for open positions within the organization. Postings include job descriptions, qualifications, and application deadlines.
 - **Applicant Tracking:** Track applicants throughout the recruitment process, including their application status (applied, shortlisted, interviewed, hired).
 - **Interview Scheduling:** Schedule interviews with applicants and send notifications to the hiring team and candidates.
 - **Interview Feedback:** Record interview feedback, evaluate applicants, and decide on the next steps (e.g., moving forward with an offer or rejection).
 - **Applicant Database:** Maintain a database of applicants for future reference in case new roles open up.
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6. Leave Management Module

This module helps manage employee leave requests and approvals, while also keeping track of leave balances.

Key Features:

- **Leave Requests:** Employees can submit leave requests directly through the system, specifying the type of leave (vacation, sick leave, maternity leave).

- **Leave Approval Workflow:** Managers can review and approve/reject leave requests based on department policies and current workload.
 - **Leave Balance Tracking:** Automatically track how much leave each employee has taken and their remaining balance.
 - **Leave History:** View a history of all past leaves taken by an employee, along with the reasons for leave.
 - **Holiday and Absence Reports:** Generate reports on the number of employees on leave during any given period, helping managers plan for coverage and staffing needs.
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7. Reporting Module

The reporting module provides insights and data analysis on various HR activities, helping management make data-driven decisions.

Key Features:

- **Employee Performance Reports:** Analyze employee performance across teams and departments, highlighting trends or discrepancies.
 - **Payroll Reports:** Generate detailed payroll reports, including total salaries paid, deductions made, and bonuses awarded.
 - **Leave Reports:** View reports on the number of leaves taken, absenteeism rates, and trends in employee attendance.
 - **Recruitment Reports:** Analyze the recruitment pipeline, including the number of applicants, interview success rates, and open positions.
 - **Customizable Reports:** HR teams can create custom reports to suit specific needs, filtering data by time period, department, or role.
 - **Exportable Data:** Reports can be exported into various formats (e.g., CSV, PDF) for further analysis or sharing with management.
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8. Notification Management Module

This module ensures communication between the HR department and employees, sending notifications about important updates or changes.

Key Features:

- **Task Notifications:** Notify employees about assigned tasks or upcoming deadlines.
 - **Leave Notifications:** Send automatic notifications for leave approvals or rejections.
 - **Payroll Notifications:** Notify employees when their salary has been processed or if there are discrepancies.
 - **General Announcements:** HR or management can send announcements to all employees, such as company-wide updates or policy changes.
 - **Custom Alerts:** Set up custom notifications for specific events, like performance review reminders or training sessions.
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9. Task Management Module

This module assigns and tracks tasks given to employees, ensuring effective workload management.

Key Features:

- **Task Assignment:** Assign tasks to employees based on their role or department.
 - **Task Tracking:** Monitor the progress of tasks and deadlines.
 - **Task Priority:** Set priority levels for tasks, ensuring important work is completed on time.
 - **Task Notifications:** Send reminders to employees about upcoming or overdue tasks.
 - **Task Reports:** Generate reports to see how many tasks have been completed, pending, or overdue for each employee.
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10. Contract Management Module

This module handles all employment contracts, ensuring they are properly maintained and updated.

Key Features:

- **Contract Creation:** HR teams can create new employment contracts, specifying details such as start date, role, salary, and terms of employment.
 - **Contract Updates:** Modify contract terms when there are changes in the employee's position, salary, or role.
 - **Contract Renewals:** Automatically send reminders for contract renewals when the contract term is nearing its end.
 - **Contract History:** Maintain a record of all past contracts and any amendments made.
 - **Contract Expiry Notifications:** Notify HR and employees when contracts are about to expire.
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These detailed module descriptions provide a comprehensive understanding of the functionality in each part of the **HRMS**. They highlight key operations that can be managed efficiently within the system using Core Java, Collections, and Hibernate as ORM

Detailed workflow

1. Employee Management Workflow

Objective: Manage employee data, including adding new employees, updating information, and handling contract details.

Steps:

1. **Employee Creation:**

- HR creates a new employee profile by filling in personal information, job role, and department.
- The system assigns the employee a unique employee ID.
- HR uploads necessary documents (e.g., offer letter, contract).
- The system stores employee data and links them to the corresponding department and role.

2. **Contract Management:**

- HR enters the employment contract details (start date, salary, role).
- The contract is saved and linked to the employee profile.
- The system automatically sets reminders for contract expiration or renewal.

3. **Profile Update:**

- Employee or HR can update employee information (e.g., address change, role change).
- Any updates trigger an automatic update across relevant modules (e.g., payroll, attendance).

4. **Onboarding/Offboarding:**

- When a new employee joins, an onboarding checklist is generated (e.g., document verification, assigning tasks).
- For exiting employees, HR initiates the offboarding process, updating the system to mark the employee as inactive.

2. Attendance Tracking Workflow

Objective: Monitor employee attendance, record working hours, and manage leaves.

Steps:

1. **Attendance Recording:**

- Employees check in and check out using an attendance system (biometric, login, or mobile app).
- Attendance is automatically recorded and linked to the employee profile.
- The system calculates work hours and tracks any overtime or absence.

2. **Leave Request:**

- Employees request leave by filling out the leave type (e.g., vacation, sick leave) and dates.
- The system checks the employee's leave balance and forwards the request to the manager.

3. **Leave Approval:**

- The manager reviews the leave request and approves/rejects it.
- Upon approval, the system updates the leave balance and attendance record.
- The employee is notified of the approval/rejection via the notification system.

4. Leave Reports:

- HR generates reports on employee attendance, absenteeism, and leaves for performance and planning purposes.

3. Payroll Management Workflow

Objective: Automate salary calculation, generate pay slips, and manage payroll-related data.

Steps:

1. Salary Calculation:

- The system calculates monthly salaries based on employee attendance, role, and contract details.
- Deductions (e.g., taxes, benefits) and bonuses (e.g., performance incentives) are automatically applied.

2. Pay Slip Generation:

- The system generates pay slips for each employee with a detailed breakdown of earnings, deductions, and bonuses.
- Pay slips are sent to employees via email or the HRMS portal.

3. Salary Payment:

- HR initiates salary payments by integrating with the company's banking system.
- Payments are processed and transferred to employees' bank accounts.

4. Payroll Reporting:

- The system generates payroll reports showing total salaries, bonuses, deductions, and tax withholdings for a given period.
- HR and finance teams use these reports for financial planning and tax compliance.

4. Performance Management Workflow

Objective: Conduct performance reviews, provide feedback, and manage promotions.

Steps:

1. Performance Review Setup:

- HR schedules performance review periods (quarterly, annual).
- Managers are assigned to evaluate employees based on set goals and criteria.

2. Employee Review:

- Managers fill out performance evaluation forms for their team members, rating them on various metrics (e.g., quality of work, teamwork).
- Feedback is recorded, and areas for improvement are identified.

3. **Feedback and Promotion:**

- The system stores feedback, and managers can discuss it with employees.
- Based on the evaluation, HR or management decides on promotions or role changes.
- Updated roles or salary increments are reflected in the employee's profile and payroll.

4. **Performance Reporting:**

- HR generates reports on overall employee performance to identify top performers and those in need of training or support.
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5. **Recruitment Workflow**

Objective: Post job openings, manage applicants, and conduct interviews.

Steps:

1. **Job Posting:**

- HR creates job postings, specifying the job description, requirements, and deadlines.
- The system publishes the job on the company's portal and external job platforms.

2. **Applicant Submission:**

- Applicants submit their resumes and cover letters through the system.
- The system assigns a unique ID to each applicant and stores their details in the applicant database.

3. **Application Review:**

- HR or the hiring manager reviews applications and shortlists candidates based on qualifications.
- Shortlisted candidates are moved to the interview stage.

4. **Interview Scheduling:**

- HR schedules interviews for shortlisted candidates, notifying the hiring team and applicants via email.
- Interview results are recorded in the system, including feedback and ratings.

5. **Hiring Decision:**

- HR marks candidates as "Hired" or "Rejected" based on interview results.
 - The system updates the recruitment status and generates an offer letter for hired candidates.
 - The hired applicant's details are transferred to the employee management module for onboarding.
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6. **Leave Management Workflow**

Objective: Handle leave requests and approvals for employees.

Steps:

1. **Leave Request Submission:**

- Employee logs into the system and submits a leave request with start and end dates and reason for leave.

- The system checks the employee's remaining leave balance.

2. Leave Approval:

- The request is forwarded to the employee's manager for approval.
- The manager approves or rejects the request based on current workload and team needs.

3. Leave Notification:

- The system sends a notification to the employee about the approval or rejection.
- Approved leave is marked on the employee's attendance calendar.

4. Leave Balance Update:

- Upon approval, the system updates the employee's leave balance and attendance record.
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7. Reporting Workflow

Objective: Generate and view various HR-related reports.

Steps:

1. Report Selection:

- HR selects the type of report they want to generate (e.g., payroll, performance, attendance, leave).

2. Data Filtering:

- Filters are applied (e.g., by department, date range, or employee role).
- The system retrieves the relevant data from the database.

3. Report Generation:

- The system generates a report and presents it in the desired format (e.g., tabular, graphical).
 - HR can download or share the report with relevant stakeholders.
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8. Notification Management Workflow

Objective: Send notifications to employees regarding important updates and events.

Steps:

1. Event Trigger:

- A trigger (such as a leave request approval, task assignment, or payroll processing) activates the notification system.

2. Notification Creation:

- The system generates the appropriate notification message, such as "Leave Request Approved" or "New Task Assigned."

3. Notification Delivery:

- The system sends the notification via email or in-app notification.
- Employees receive the notification and can take action if necessary (e.g., view details, acknowledge receipt).

9. Task Management Workflow

Objective: Assign tasks and monitor their progress.

Steps:

1. **Task Creation:**

- Managers create tasks and assign them to employees, setting deadlines and priority levels.
- The system saves the task in the employee's task list.

2. **Task Progress Tracking:**

- Employees mark tasks as "In Progress" or "Completed" in the system.
- The system updates the task status and sends notifications to the manager.

3. **Task Completion:**

- Once a task is marked as completed, the system notifies the manager for review.
- The manager marks the task as approved, and the system archives it.

4. **Task Reports:**

- The system generates task reports showing pending and completed tasks for each employee.
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10. Contract Management Workflow

Objective: Manage employee contracts, renewals, and updates.

Steps:

1. **Contract Creation:**

- HR creates a new contract for a new hire, entering the terms (e.g., role, salary, duration).
- The system links the contract to the employee's profile.

2. **Contract Updates:**

- For contract renewals or updates (e.g., salary increment), HR modifies the contract in the system.
- The system records the changes and updates the employee's payroll data if applicable.

3. **Contract Expiry Notification:**

- The system sends reminders to HR and the employee before the contract expiry date for renewal or termination actions.

4. **Contract Archive:**

- Expired contracts are archived, but remain accessible for historical reference.
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This detailed workflow ensures that the HRMS runs smoothly, with each module integrated to provide efficient management of HR operations using Core Java, Collections, and Hibernate as an ORM tool.